Worksheet 1: Lipid Digestion and Fair Comparisons — Answers

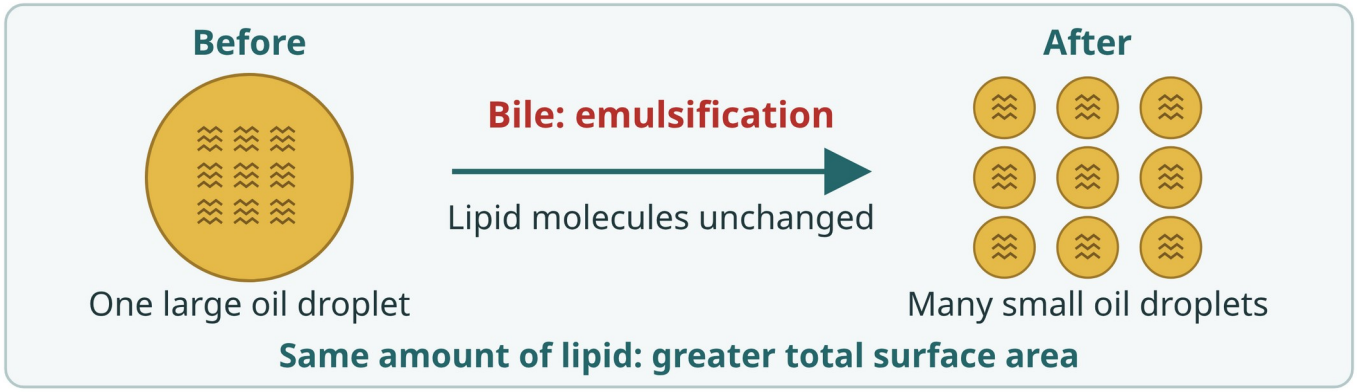
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Read the topic notes, then answer using the context and model results.

Topic notes

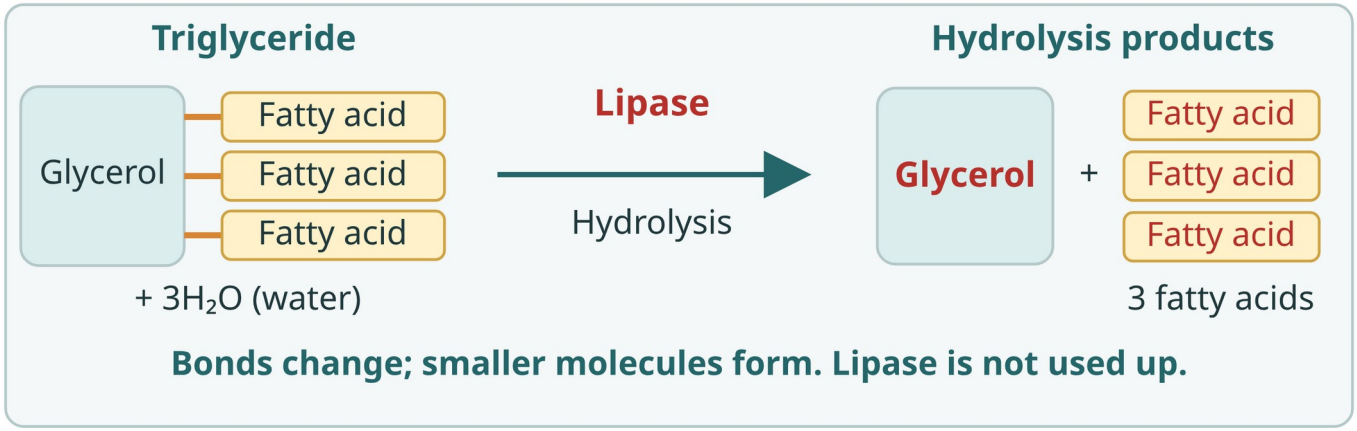
01 Bile: emulsifying lipids

In this model, X is **bile**. Made in the liver and stored in the gall bladder, **bile** contains no digestive enzymes. It **emulsifies** large oil droplets into **small droplets**, increasing the **total surface area** for **lipase** to act on. This is **physical digestion**: lipid molecules remain unchanged.



02 Lipase in pancreatic juice: chemical digestion

In this model, Y is **pancreatic juice** containing **lipase**. **Lipase** catalyses **chemical digestion** of lipids. This course uses a simplified complete-hydrolysis model: **triglycerides** form smaller **glycerol** and **fatty-acid** molecules for absorption. **Lipase** is a **protein**; high heat can **denature** it and stop it working.



Part 1 | Context and fair comparisons

Researchers add digestive juices X and Y to oil–water mixtures, shake them, and compare their appearance.



Research question: How do combinations of digestive juices affect the appearance of an oil–water mixture?

1. Identify the variables

Circle the appropriate factors in each row. You may choose more than one.

Variable type	Factors to choose from
Independent variable Factor changed	Total volume of oil and water / Reaction temperature / Reaction time Total volume of liquid added / Method of shaking / Appearance of the mixture / **Combination of digestive juices**
Dependent variable Outcome measured	Total volume of oil and water / Reaction temperature / Reaction time Total volume of liquid added / Method of shaking / **Appearance of the mixture** / Combination of digestive juices
Controlled variables Kept the same	**Total volume of oil and water** / **Reaction temperature** / **Reaction time** **Total volume of liquid added** / **Method of shaking** / Appearance of the mixture / Combination of digestive juices

2. Select the assumptions for a fair comparison (choose all that apply)

(An assumption is a condition we have not proved but take as true, to help make the experiment results valid.)

☑ **All sets start with the same amounts of oil and water, and receive the same total volume of added liquid.**

☐ The X + Y set may receive more total liquid to compare it with the other sets.

☑ **All sets are shaken in the same way.**

☑ **All sets are observed after the same reaction time.**

3. Design a control group

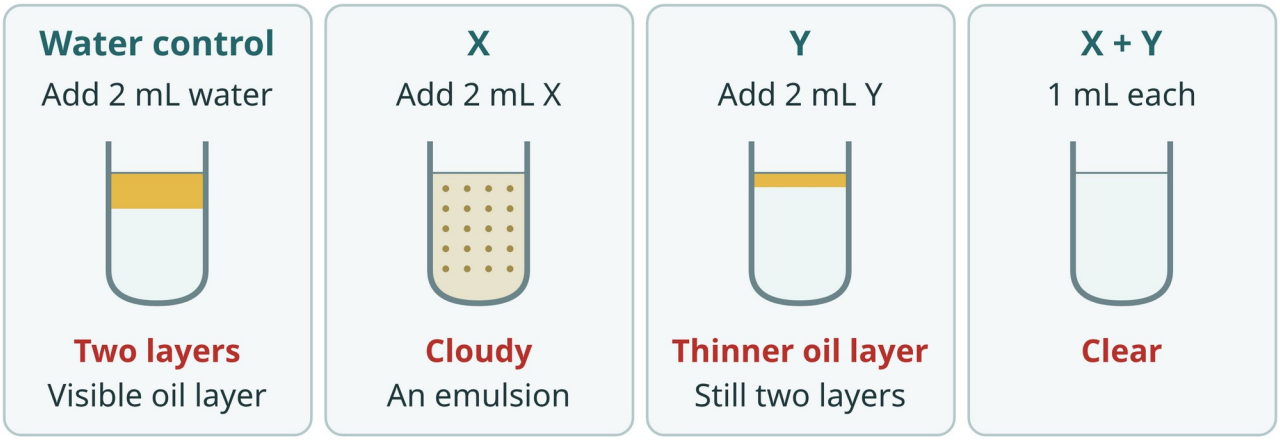
Does this experiment need a **control group**? If so, how would you design the comparison setup?

(Hint: A control group provides a starting point for comparison. It helps you check that changes are caused by the independent variable, rather than other outside factors. The control group therefore does not receive the treatment being tested.)

Yes. Add 2 mL of water to an oil–water mixture as a control and compare it with X, Y and X + Y. Keep starting oil/water amounts, total added volume, temperature, time and shaking the same.

Part 2.1 | Analyse the four basic experiments

These are model results, not tests for digestion products. All sets use the same conditions and observation time.



4. What does digestive juice X do?

Digestive juice X mainly causes lipids to form a (**cloudy emulsion** / clear solution).

5. How does digestive juice X help Y act?

X disperses large oil droplets into **small droplets**. Their **total surface area** (**increases** / decreases), helping Y act.

6. What can this experiment tell us?

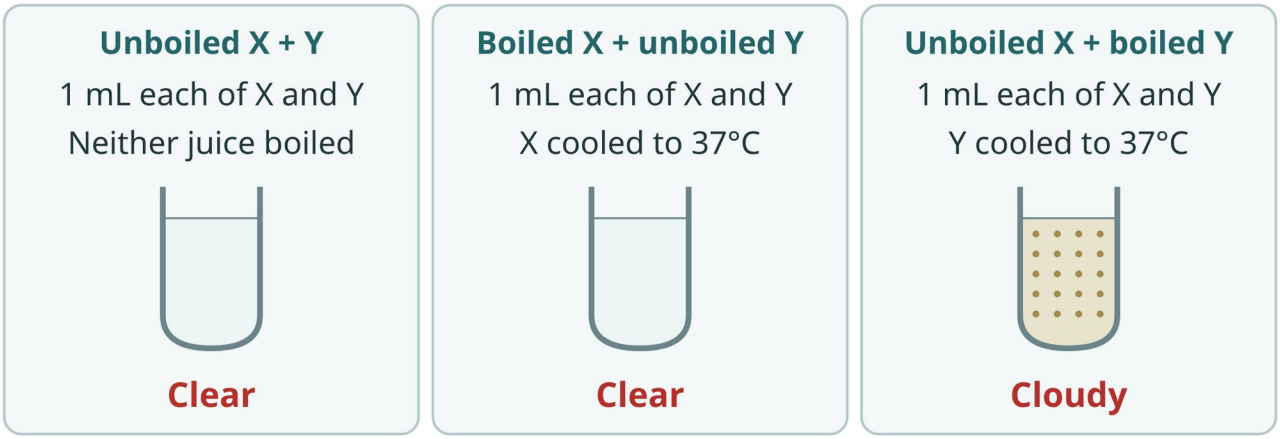
We observed only the oil layer, cloudiness or clarity, without testing digestion products. Which statement is most appropriate?

✓ A. Appearance gives clues about the juices, but alone cannot prove that glycerol and fatty acids have formed.

B. A clear mixture fully proves that lipids have broken down into **glycerol** and **fatty acids**.

Part 2.2 | Effects of heating the juices

Use unboiled X + Y as the baseline. Boil only one juice at a time, then cool it to 37°C. Add 1 mL each of X and Y to every set; keep other conditions the same.



7. Extension conclusion

The juice that loses its action after boiling is (digestive juice X / **digestive juice Y**).

Part 3 | Integrated explanation

8. Using the experimental results and topic notes, explain how bile and lipase in pancreatic juice work together to help digest lipids, and how boiling affects their actions.

Bile contains no digestive enzymes. It physically emulsifies large oil droplets into small droplets without changing the lipid molecules. The greater total surface area lets lipase contact more lipid. Lipase in pancreatic juice catalyses chemical digestion, producing glycerol and fatty acids in the complete-hydrolysis model used in this course. Lipase is a protein. Boiling denatures it and stops its catalytic action; cooling does not restore it. Bile retains its emulsifying action after boiling in this teaching model.